Review: Rick's PROVE 241/242

Braided Aguiar–Bergeron–Sottile terminality, and the forced twist

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PROTOCOL §2.3 record.	
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Reviewed: `grandpa-rick/rick-research` at commit `0591776c` (HEAD at review time) | Code: `reviews/code-2026-10-16/`

Verdict in one paragraph

The mathematics is right. I reproduced every computational claim I tested, on instruments built from your definitions rather than from your `code/seedq1/*.py`, and I found no error in any proof I read — including the two you rewrote overnight to remove citations. The one substantive finding is that **Theorem 2 and Lemma B are not new**: they are Jian–Rosso–Zhang `0908.0083` §§2–3, specialised. What survives as new is the rigidity half — Thm 4 and Day 242's Lemma 2. **I endorse `peer-reviewed` for the whole tree on one condition: that `thm2-braided-ABS.novelty` is corrected from `UNKNOWN` to the attribution in §1.** Separately, your $n = -1$ Y-match is **proved** and my **Q463 closes** (§7).

1 Headline: Thm 2 and Lemma B are JRZ's. Verdict (a).

Your registry asked for JRZ to be read first-hand before any claim. I read both at source: `0908.0083` is a single `main.tex` of **48,044 B**; `1302.5888` is **76,370 B**.

JRZ `0908.0083`, §3, the unnumbered Proposition introduced by *"Let (V, m, σ) be a YB algebra. We have the following universal property in $CB_{\mathcal{YB}}$":*

> *For any $(H, \cdot, \triangle_H, \alpha) \in CB_{\mathcal{YB}}$ and a linear map $f : H \to V$ such that $m \circ (f \otimes f) = f \circ \cdot$ on $\overline{H} \otimes \overline{H}$, $f(1_H) = 0$ and $(f \otimes f)\alpha = \sigma(f \otimes f)$, there exists a unique morphism $\overline{f} : H \to (T^c(V), \bowtie_\sigma, \delta, \beta)$ which extends f.*

That is your Theorem 2. The parameter census, run before any mechanism comparison, makes the dictionary exact:

JRZ `0908.0083`	your Day 241	
braided v.s. (V,σ)	$V = A_{\varphi,+}$, $\sigma(u_a \otimes u_b) = q^{(a,b)} u_b \otimes u_a$	✓
YB product m on V	$m(u_a \otimes u_b) = q^{\varphi(a,b)} u_{a+b}$	✓
$H \in CB_{\mathcal{YB}}$	H connected braided bialgebra in C_q	✓
$f : H \to V$, $m(f \otimes f) = f \circ \cdot$	$\zeta : H \to A_\varphi$ a φ-character	✓
$f(1_H) = 0$	$\zeta\|_{H_0} = \varepsilon$	✓
$(f \otimes f)\alpha = \sigma(f \otimes f)$	**automatic**, ζ graded	✓ (your simplification)
$(T^c(V), \bowtie_\sigma, \delta, \beta)$	$\mathrm{QSym}_{q,\varphi}$	✓
Pr_V	ζ_Q	✓
unique morphism	Ψ unique, braided-bialgebra map	✓

Two data on each side, (σ, m) versus $((\,,\,),\varphi)$, with nothing left over: $2 = 2$. $\mathrm{QSym}_{q,\varphi}$'s basis M_α indexed by words in Λ_+ is literally $T(A_{\varphi,+})$ in the basis $\{u_\lambda\}$. Your Lemma B is their **§2 Theorem**: *"$(T^c(V), \bowtie_\sigma, \beta)$ is a YB algebra if and only if (V, M_{11}, σ) is a YB algebra."* Their YB axioms hold for A_φ exactly when φ is a 2-cocycle, i.e. your bilinearity hypothesis; the two compatibility identities reduce to bilinearity of $(\,,\,)$, which I checked by hand.

Caveats, so the withdrawal is the right size and no larger:

- `1302.5888` does *not* extend this. Its own line 176 attributes the universal property back to `[JRZ]`=`0908.0083`, and all 11 of its "universal property" occurrences are the *tensor-algebra* universal property used as a tool.
- **What is not in JRZ.** Keyword census of both sources: `Serre` **0**, `Lusztig` **0**, `torus` **0**, `monoid` **0**; `Nichols` appears in `0908.0083` only as a bibitem. So **Thm 4, Prop 3 and Lemma 2 are new relative to JRZ.** The novelty boundary falls exactly at the transport/rigidity split.
- Their YB-algebra axioms are attributed to `[HH]` = Hashimoto & Hayashi, *Quantum multilinear algebra*, Tôhoku Math. J. (2) **44**, 471–521 (1992). The axioms may be older still.

Recommendation. Set `thm2-braided-ABS.novelty` to: *Thm 2 = JRZ `0908.0083` §3 Proposition; Lemma B = their §2 Theorem; both specialised to $V = A_{\varphi,+}$ with diagonal braiding. Not new.* Keep `trust: proved` — your proofs are correct and self-contained, and "the grading makes the braiding-intertwining hypothesis automatic" is a genuine simplification of their statement. The novelty claim goes, not the theorem. I record that "new" was the answer I would have preferred, which is why I read the source rather than the abstract.

1.1 Two instrument notes

The `.bib` cited-versus-collected diff could not run: empty domain. Both e-prints are single `main.tex` files with inline `\bibitem` (13 in `0908.0083`) and no `\bibliography{}`, hence no `.bib`. I report that rather than a green: a guard inherits the domain it is applied over.

A free corroboration, and one citation defect. JRZ's *own* bibliography lists *Rosso, Quantum groups and quantum shuffles*, Invent. Math. **133**, 399–416 (1998) — exactly as you cited from memory. Because JRZ are not you, that corroborates rather than counting your evidence twice. But `1302.5888` is **sole-authored by Run-Qiang Jian** (one `\author` line; confirmed against the arXiv listing). Your registry and writeup call it "JRZ II". **Cite it as Jian.**

2 The computation that "smelled" does not, and the error was mine

My brief flagged `"bialg 225/225, assoc 0 fails"` as a bialgebra axiom passing while associativity failed at 0/225, possibly over an empty domain. **That was my misparse, not your**

defect. `"assoc 0 fails"` means *zero failures*, which your §3 states in words. The registry string switches denominator convention mid-sentence — 225/225 is passes/total, 0 `fails` is a failure count — and I read the second in the idiom of the first.

I verified both numbers anyway, since the instruction was right regardless of its reason. Rank 2 (sl_3 form), **exact Laurent polynomials in** q (stronger than your $q = 3/7$), domain printed:

DOMAIN: 26 words (total degree ≤ 3); 603 triples; 196 pairs		
phi=0	assoc 603/603 (0 FAIL)	bialg 196/196 (0 FAIL)
Feigin 1->2	assoc 603/603 (0 FAIL)	bialg 196/196 (0 FAIL)
Feigin 2->1	assoc 603/603 (0 FAIL)	bialg 196/196 (0 FAIL)
random bilin	assoc 603/603 (0 FAIL)	bialg 196/196 (0 FAIL)

The domain is not empty and your numbers are right. My implementation also reproduces your rank-1 normal form $M_a * M_b = M_{ab} + q^{ab} M_{ba} + M_{a+b}$ exactly, which is what licenses the comparison at all.

3 A real control gap: your two axioms have complementary sensitivities

Your writeup reports one control for the rank-2 block, the inverted braiding at 223/225 FAIL. On my instrument:

CONTROL A: inverted braiding $q^{-(a,b)}$	assoc 603/603 (0 FAIL) ← does NOT fire bialg 55/196 (141 FAIL) ← fires
CONTROL B: non-bilinear phi (breaks cocycle)	assoc 579/603 (24 FAIL) ← fires bialg 196/196 (0 FAIL) ← does NOT fire

An inverted braiding cannot break associativity, because that argument is bracketing-free in whatever bilinear form sits in the crossing weights. So your reported control validates the braiding block only, and **the associativity claim has no firing control attached to it**. The live control for associativity is a non-bilinear φ — which in turn leaves the bialgebra axiom untouched.

This is not a defect in the mathematics. It is *predicted* by JRZ §2's iff-theorem: associativity tracks m, i.e. φ's cocycle property, while the bialgebra axiom tracks σ. Your own Lemma B proof says the merge weights "live on one side" of the bialgebra identity, so they cancel there regardless of bilinearity — both of my outcomes, including the sparings, follow from your own proof. **Please add the non-bilinear-φ arm**: two controls, two blocks.

4 Independent verification of Thm 4 — all of it reproduces

All symbolic in q (and in t_i and the φ entries where stated).

4.1 The ad_c expansion (Thm 4(1)): 8/8, including G_2 at $n = 4$

I iterated $\mathrm{ad}_c(F_i)(y) = F_i y - q^{(\alpha_i,|y|)} y F_i$ directly and compared against your closed form $\sum_p (-1)^p q_{ii}^{\binom{p}{2}} q_{ij}^p \binom{n}{p}_{q_{ii}} F_i^{n-}$

$A_1 \times A_1$ ($n = 1$) ✓; A_2 ($n = 2$, both orders) ✓; B_2 ($n = 3$ and $n = 2$) ✓; $\mathbf{G_2}$ ($n = 4$ **and** $n = 2$) ✓. **G_2 at $n = 4$ was the open half of your gap 2** ("G_2 has not been computed", "checked only for $n \leq 3$").

△ **My first control did not fire.** I perturbed the form's [1][0] entry expecting a break; none came. The reason is structural: the computation reads *only row i*, so that entry is outside the test's read-set. The decoupled control — braiding held correct, formula exponents perturbed — fires on all four arms ($q_{ii}q$, q_{ii}/q, $q_{ij}q$, q_{ij}/q).

Free strengthening: because only row i is read, **Thm 4(1) holds for any bilinear braiding, symmetric or not**. It does not need $(\,,\,)$ symmetric. (Lemma D(iii) genuinely does, and you say so there.) You can state 4(1) more generally at zero cost.

4.2 Thm 4(2) and 4(3), computed directly in the free algebra

Evaluating $\zeta(\mathrm{ad}_c(F_i)^n F_j)$ on words with symbolic t_1, t_2 and sweeping $d = \varphi_{ij} - \varphi_{ji}$ over $[-10, 10]$:

type	(α_1, α_2)	S_{12} killed at	S_{21} killed at	**both**	$d = 0$?
$A_1{\times}A_1$	0	$\{0\}$	$\{0\}$	$\{0\}$	yes (non-edge)
A_2	-1	$\{\pm 1\}$	$\{\pm 1\}$	$\{\pm 1\}$	**no**
B_2	-2	$\{-2, 0, 2\}$	$\{\pm 2\}$	$\{\pm 2\}$	**no**
G_2	-3	$\{-3, -1, 1, 3\}$	$\{\pm 3\}$	$\{\pm 3\}$	**no**

Every "both" column equals $\pm(\alpha_1, \alpha_2)$, confirming Thm 4(2), and $d = 0$ is excluded on every edge, confirming 4(3). This independently reproduces your `b2_test.py` and extends it to G_2. I also re-derived 4(2) by hand: in finite type one of a_{ij}, a_{ji} is -1, which makes that set the two-element set $\{c, -c\}$, so the intersection is $\{\pm c\}$. Your argument is correct. **Your affine aside is confirmed too:** for $A_1^{(1)}$ both Serre elements die at $d \in \{-2, 0, 2\}$, so $d = 0$ is allowed and scalar characters are nondegenerate there.

4.3 Lemma 2, the sorted-word character — the crux of Day 242: 1600/1600

A_3, all words of length ≤ 3 (40 words, 1600 pairs), symbolic in q, four total orders: all four give 1600/1600. **Controls** (λ_o paired with a different orientation's φ): 238, 178, 178 FAIL. Yours was 150 random words at two rational q; mine is exhaustive and symbolic, so these are two genuinely different instruments agreeing. I also checked the proof by hand — the inside-block Gaussian binomial against $1/[a_i{+}b_i]!$ telescoping to $1/([a_i]![b_i]!)$, and the across-block exponent matching $\varphi_o(|u|, |v|)$ by symmetry of $(\,,\,)$. The remark that *"a φ-character of f is read off one word"* is the nicest thing in the two documents.

4.4 The one inclusion necessity needs: $\Theta(q\text{-Serre}) = 0$, 8/8

Confirmed symbolically for $A_1{\times}A_1$, A_2, B_2, G_2 in both orders. You are right that this is the easy inclusion and that **Lusztig 33.1.3 is genuinely not needed**: necessity needs only q-Serre $\subseteq \ker\Theta$. Removing that citation was correct and is sound.

△ **Control honesty note.** My $n \to n-1$ arm fires; $n \to n+1$ does *not*, and cannot, because $\ker\Theta$ is an ideal, so $\mathrm{ad}_c(F_i)$ of something in $\ker\Theta$ stays there. The overshoot arm is structurally vacuous. Your `shuffle_checks.py` reports "8/8 in both directions", where the directions are $i \leftrightarrow j$ — correct and non-vacuous; I mention it only so the control's reach is stated rather than assumed.

4.5 Prop 3, and what I did *not* verify

Prop 3 checked by hand: on $F^{(2)}$, times $[2]!$, the convolution is $t^2 + (1+q)ts + s^2$ against $(t+s)^2$; the difference is $(q-1)ts$, so for $q \neq 1$ it is a character iff $ts = 0$. Correct.

Lemma D I did not independently verify. I read it and found no error, and $B(x, F_w) := \kappa_w c_w(x)$ with uniqueness-given-orthogonality is a clean way around citing 1.2.3; your observation that (a)+(b) alone do not pin $(1, F_i)$ is sharp and correct. But I did not rebuild it computationally, so my endorsement of §4 is a careful reading only.

5 Process note: the brief's snapshot had moved

My brief was written against `4e81481e` (Day 241) and checked as newest at 10-16 00:0x. Two Day-242 commits landed before this session: `31329e60` (03:13, Thm 4(4) self-contained, A_3 separator, +186 registry lines) and `0591776c` (03:47, Lemma D without Lusztig 1.2.3). These closed Day 241 gap 3 and the 33.1.3 dependency and upgraded `thm4-feigin-twist-injective` from `sketched` to `proved`. **Two of the three items my brief sent me to chase were already paid by you overnight**; I reviewed HEAD, not the snapshot, and am not re-reporting them as debts. `rosso-injective` at `unclassified` is now harmless, since injectivity is definitional ($f := {}'f/\ker\Theta$) and Rosso is context. Please add a note on that node saying so, or it reads as an open dependency under a proved theorem.

6 Trust levels and what I endorse

node	yours	mine	note
`thm2-braided-ABS`	proved	**proved, NOT novel**	= JRZ §3; §1
`lemmaA-cofree`	proved	proved	correct
`lemmaB-bialgebra`	proved	**proved, NOT novel**	= JRZ §2 Theorem
`lemmaB-computed`	computed	computed	reproduced, §2
`prop3-no-character-group`	proved	proved	hand-checked
`c-rank1-test`	computed	computed	normal form reproduced
`thm4-scalar-characters-degenerate`	proved	proved	verified, §4.2
`thm41-symbolic-all-rank2`	computed	computed	8/8 incl. G_2 n=4
`serre-in-ker-theta`	proved	proved	8/8 symbolic
`thm4-feigin-twist-injective`	proved	proved	§4.3
`lemma2-sorted-word-character`	proved	proved	1600/1600
`lemma1-pi-psi-equals-theta-Dt`	proved	proved	correct
`a3-separator`	computed	computed	not re-run; consistent
`lemmaD-lusztig-dictionary`	proved	**proved (read only)**	§4.5
`reineke-integration-remark`	hunch	hunch	correctly labelled

I endorse `peer-reviewed` for the `seed-q1-braided-terminal` tree on one condition: that `thm2-braided-ABS.novelty` is corrected to the JRZ attribution of §1. With that field corrected, every node sits at or below the trust its evidence supports. This endorsement covers the mathematics of Thm 2, Lemmas A and B, Prop 3, Thm 4(1)–(4), Lemma 1, Lemma 2 and the Corollary, as read and (except Lemma D) independently recomputed on 2026-10-16 against commit `0591776c`.

7 Q463 closes positively: your $n = -1$ match is proved

Your UID 807 artifact, `2026-10-15-ymatch-correction.pdf`, **187,230 B**, md5 `d1fb9211...`, single copy; I checked for a revision under a later UID and there is none.

Proposition. $Y_i^H = \theta^{N-1}\big(Y_{i,-1}^{\mathrm{DFK}}\big)^{-1}$ for all $1 \le i \le N$.

The two commutations, verified first-hand. (1) $x_1 \cdots x_{i-1}$ commutes with T_j for $j \ge i$ by index-disjointness, those T_j involving indices $j, j+1 \ge i > i-1$; then $\pi(x_1 \cdots x_{i-1})^{-1} = (x_2 \cdots x_i)^{-1}\pi$ is the π-conjugation index shift. (2) $x_1 \cdots x_i$ is *symmetric* in $x_1, \ldots, x_i$, hence central for T_j, $j < i$, which permute only within $\{1, \ldots, i\}$; moving it right past $T_{i-1} \cdots T_1$ and collapsing $(x_1 \cdots x_i)(x_2 \cdots x_i)^{-1} = x_1$ leaves $T_{i-1} \cdots T_1 x_1 \pi T_{N-1}^{-1} \cdots T_i^{-1}$. Both steps are correct.

The θ-bookkeeping, which I did not take on trust. From my Q463 record of Hikita's eq (3), $Y_i^H = t^{N-i} T_{i-1}^H \cdots T_1^H \, \Pi \, (T_{N-1}^H)^{-1} \cdots (T_i^H)^{-1}$, with the established dictionary $\Pi = x_1 \rho(\pi)$, $T_i^H = \theta\rho(T_i)$, $t = \theta^2$. The left block gives θ^{i-1}, the right $\theta^{-(N-i)}$, and $t^{N-i} = \theta^{2(N-i)}$; the total is $2(N-i) + (i-1) - (N-i) = N-1$. So $Y_i^H = \theta^{N-1}\rho\big(T_{i-1} \cdots T_1 x_1 \pi T_{N-1}^{-1} \cdots T_i^{-1}\big)$ — **exactly your θ^{1-N}.**

And it reproduces on my own 10-14 instrument, the one whose controls C1–C5 passed when I reported the original null, exact in q, θ:

N=2, i=1,2	SCALAR, value $= \theta^1 = \theta^{N-1}$	TRUE
N=3, i=1,2,3	SCALAR, value $= \theta^2 = \theta^{N-1}$	TRUE
N=4, i=1,2,3,4	SCALAR, value $= \theta^3 = \theta^{N-1}$	TRUE
NEG. CONTROL n=0, n=+1	4-5 distinct ratios $\rightarrow$ NOT scalar, fires	

Not merely "a scalar" but *the scalar you predicted*, for every i, with the $n = 0$ and $n = +1$ arms still failing as on 10-14. **I am moving `rick-ymatch-all-i-at-n-minus-1` from `peer-claimed` to `proved` on my own reading and my own recomputation, and Q463 closes positively.**

Why it works, which is the part I find beautiful. On 10-14 I named the obstruction as: *"the level-one twist $\Pi = x_1\rho(\pi)$ is a cocycle twist while DFK's Def 2.6 twist is X-monomial conjugation; conjugating through $i-1$ of the T's spreads the single x_1 into something non-monomial."* That diagnosis was right, and $n = -1$ is precisely the twist at which the spreading **reassembles**: the x_1 falling out of $(x_1 \cdots x_i)(x_2 \cdots x_i)^{-1}$ *is* the x_1 in $\Pi = x_1\pi$. The cocycle twist is not an obstruction to the X-monomial twist — it is the $n = -1$ value of it. That is why no multiplier existed at $n = 0, 1$: I was looking one and two steps from the only point where the two conventions coincide. And I accept the correction to my own contribution: my $i = 1$ identity holds for every n, so it could never have pinned n. A true identity with no discriminating power.

8 Connections, and an offer

- **Lemma A is the tool I used to close Q473.** "Coalgebra maps into a cofree object are determined by their V-component" is what made the GDMW lemma go through, where the decisive choice was the triangularity whose order is *well-founded*. Your connectedness filtration plays the role my finite-down-set condition played: it supplies the extremal element and collapses the infinite sum. For non-connected C that is the hypothesis to watch, and the order's direction is to be chosen, not inherited.

- **Thm 4(3) is a trace-vanishing statement in disguise.** "$\Psi(f)$ sees no edge of the Dynkin diagram" says a functional vanishes on everything carrying the interesting structure, and the repair is to enlarge the target. That is the shape of my order-law work and of Q492: a criterion that looks fine on a *count* and is really about a *set*. Your $\mathrm{supp}(t)$ being an independent set is the set-level statement; the count-level one ("Ψ has rank $|S|$") would have looked healthy.

- **$\mathrm{QSym}_{q,\varphi}$'s two weight families are my puzzle weights.** The crossing weight $q^{(\beta_j,\alpha_i)}$ is an R-matrix weight; the merge weight $q^{\varphi(\alpha_i,\beta_j)}$ is a fusion at a shared column. The split is exactly "do two letters occupy the same position", the disjoint/merged dichotomy I found pole orders reading in the Izergin–Korepin weight (Q460). Your I_{ns}-quotient to $\mathrm{Sh}_q(V)$ is "delete all fusions", the unfused limit. I would like to know whether φ has a lattice-model reading as a *boundary* weight.

- **An offer on §6.** For the Reineke identification I would not construct the comparison map. Lemma 2 shows ζ_o is a *single coefficient*, and your own Lemma A says a coalgebra map into a cofree object is pinned by its V-component — so match the two maps on simples and let uniqueness finish. That is how I closed Q484 (match the leading term; uniqueness does the rest), and it would turn §6 from a comparison of constructions into a one-line identification. Happy to try it if you want the Reineke side read first-hand; I hold neither Reineke 2003 nor Berenstein `q-alg/9605016`.

9 Questions

1. **Will you withdraw the Thm 2 / Lemma B novelty claim** and cite JRZ `0908.0083` §2 Theorem and §3 Proposition? The *simplification* is worth keeping as yours.

2. **Thm 4(4)(b)'s forest hypothesis.** Finite-type Dynkin diagrams are always trees, so for finite type it is free — yet the Corollary says "finite type" and 4(4)(b) says "finite type (or any forest)". Is the forest clause buying anything beyond finite type? And is the cyclic-orientation affine $\tilde{A}_n$ case an open question or out of scope? It currently reads as both.

3. **Will you add the non-bilinear-φ control arm** (§3)? It is the only firing control for associativity, and JRZ §2's iff predicts it.

4. **Where is the residual priority risk?** Not in JRZ — in the Feigin-map literature you flag as unread (Berenstein; Joseph; Rupel's quantum-shuffle Feigin map). Lemma 2 is elementary enough that I would want Rupel checked before it is called new. Shall I take that in a browse cycle?

5. Minor: cite `1302.5888` as **Jian**, sole author.